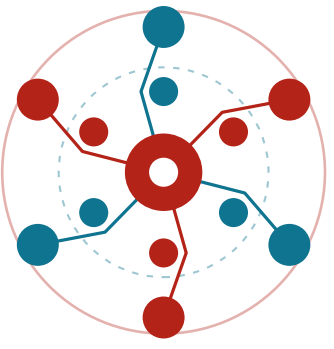
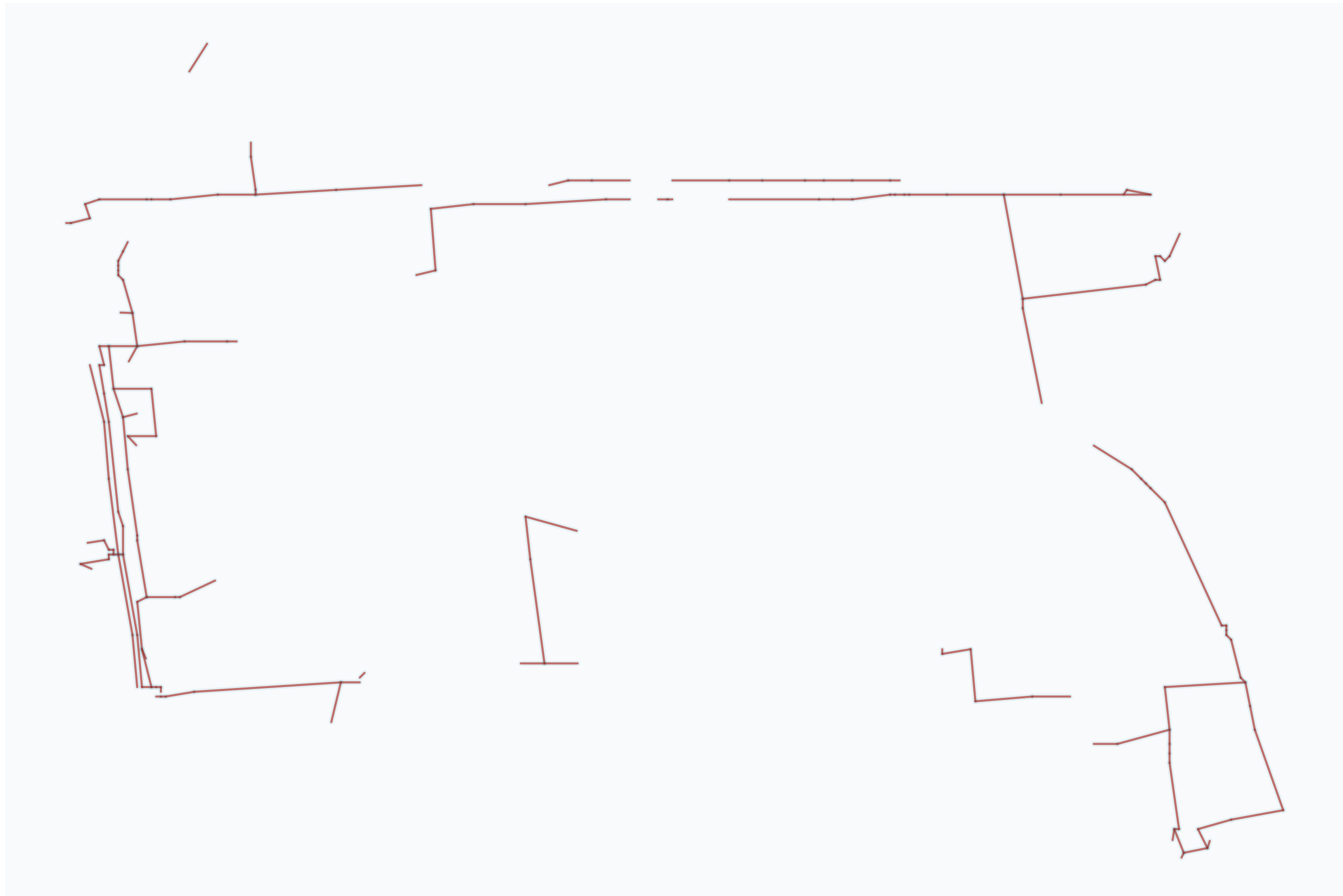




Technical route (Phase 0 → deliverables)

Real Physarum 167-edge skeleton (method validation)



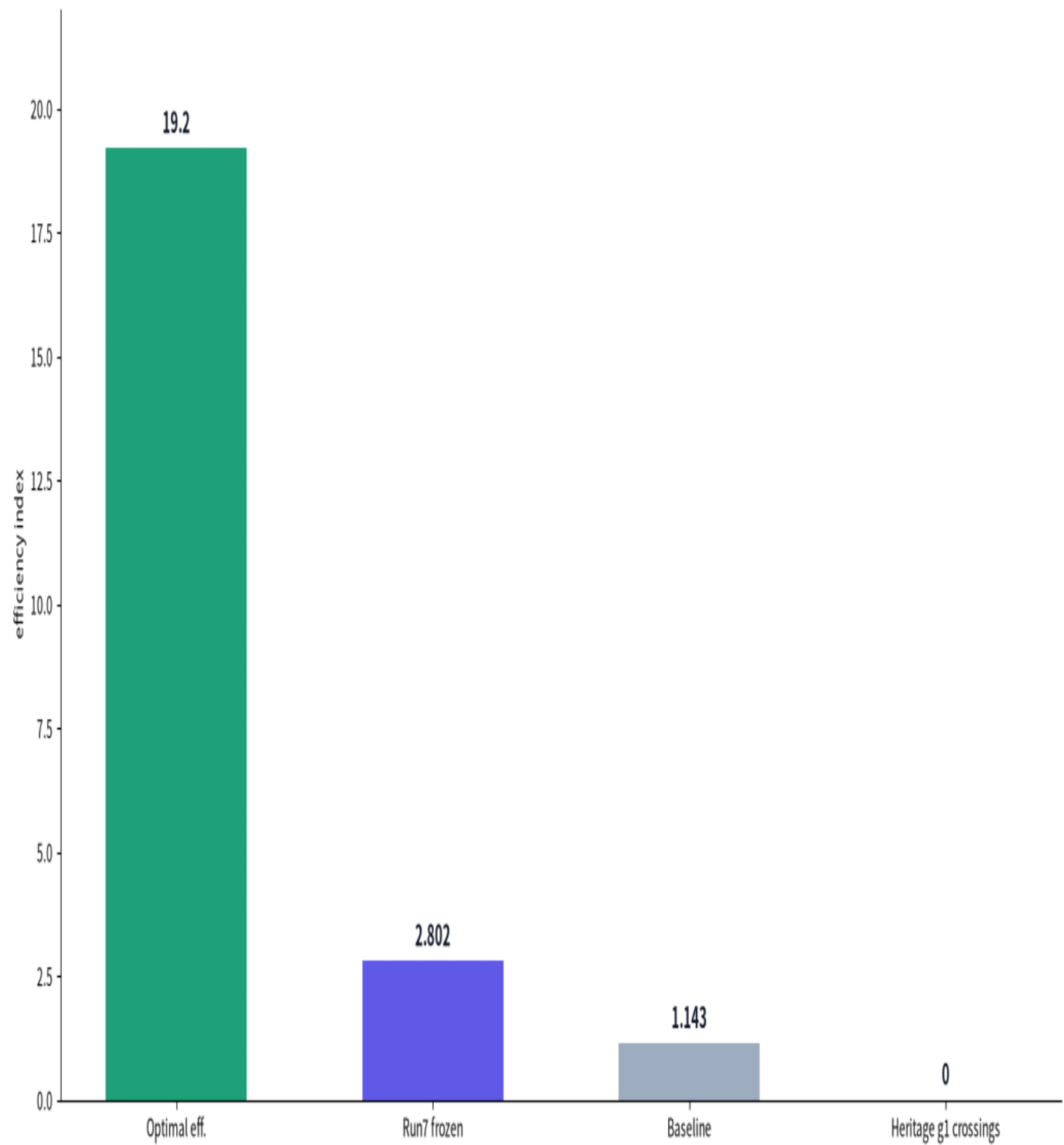
★ 黏面效率 1.143 → 19.20 (约 16.8×)

★ 主方案 Plan03 城市融合 UDS 80.34



Core metrics & method validation

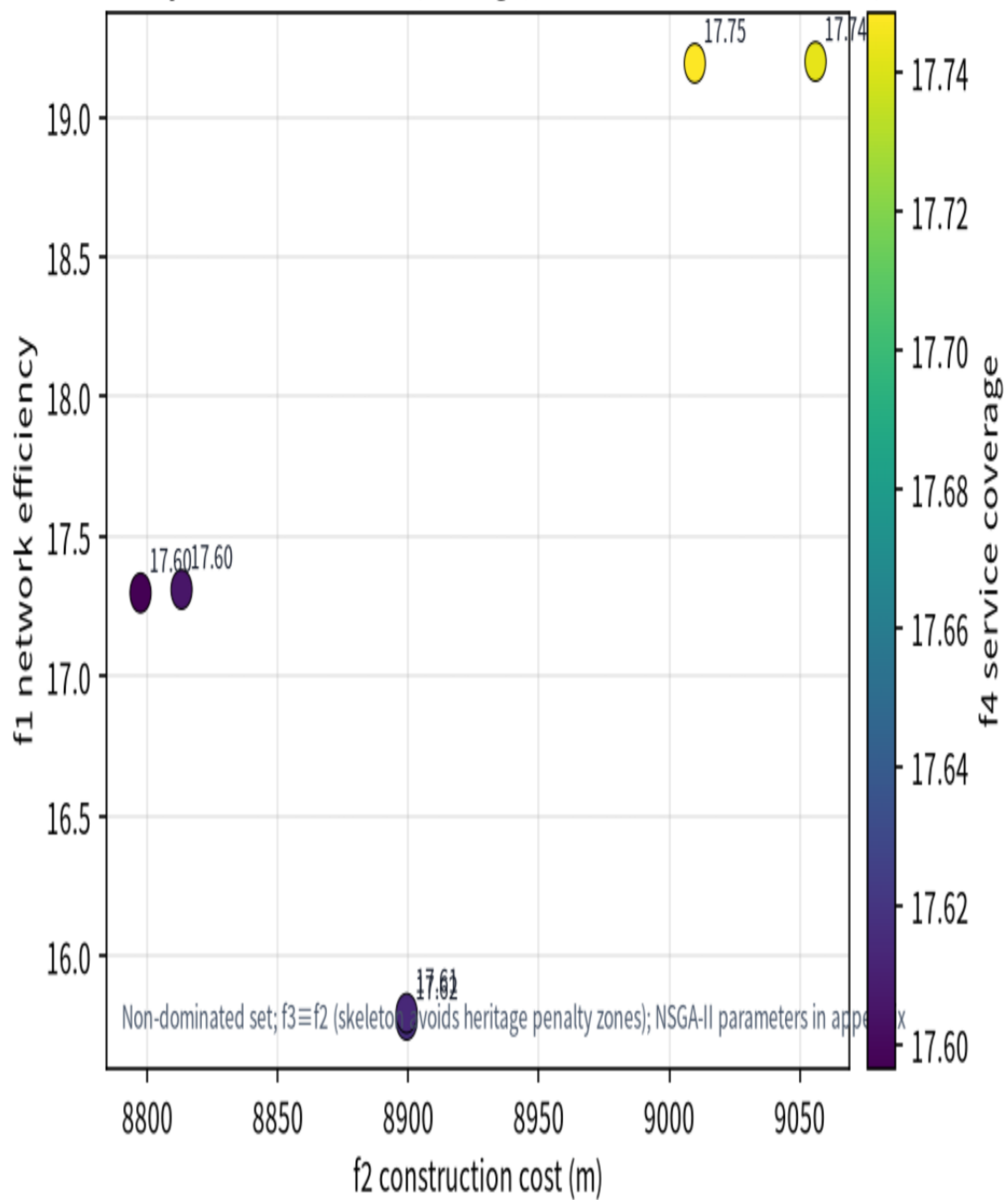
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective f3≡f2 degenerates to cost — "0 heritage crossings" is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

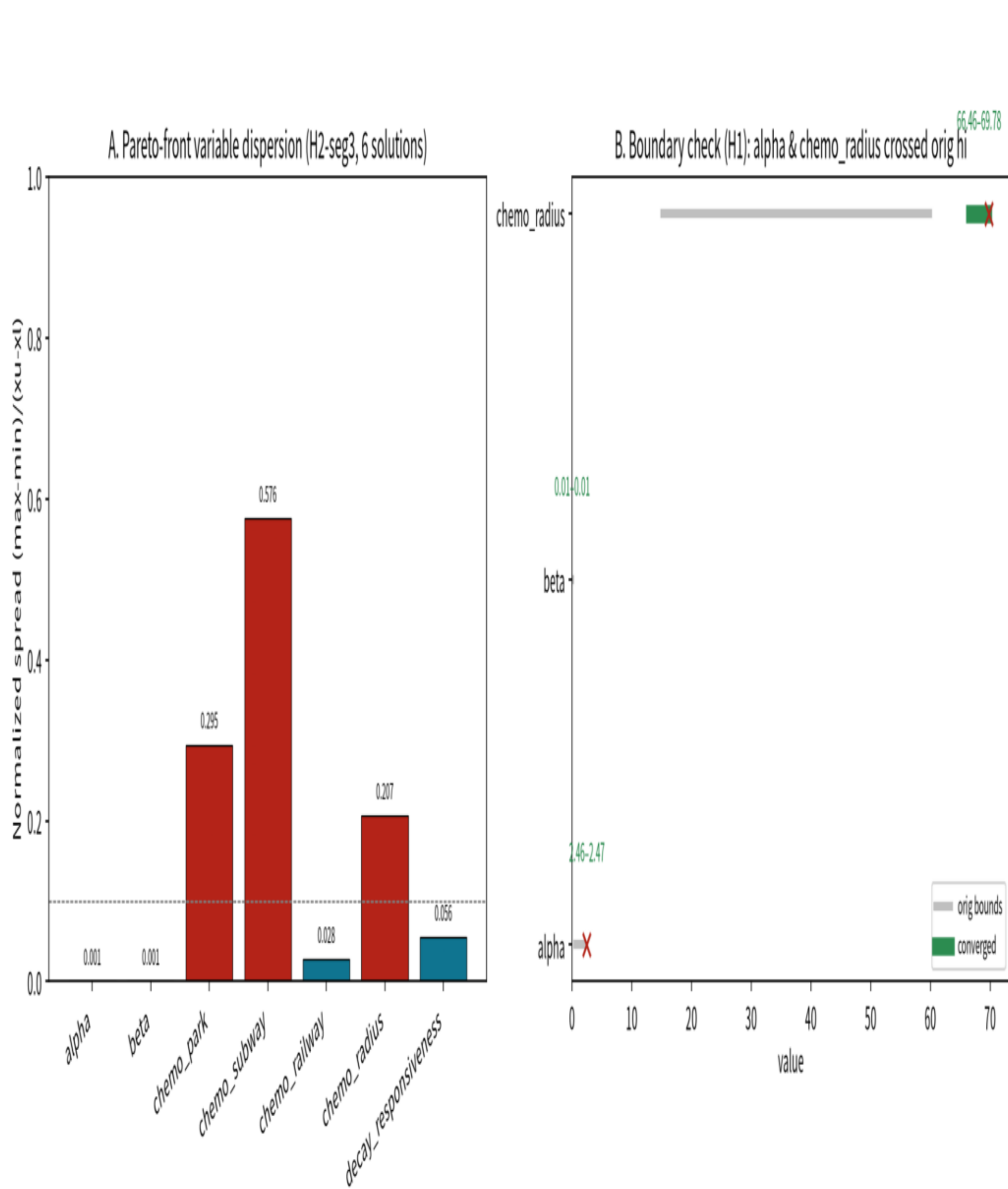
Four-objective Pareto front (6 sol.)

Four-objective Pareto front (H2-seg3, 6 non-dominated solutions)



Parameter sensitivity (suggestive)

Parameter sensitivity – Bio-Physarum + NSGA-II (real runs, method validation)



Suggestive phasing gantt (6 projects × 3 phases)

Suggestive phasing gantt (6 projects x 3 phases)



19.20

Optimal eff.

1.143

Baseline

2.802

Run7 frozen

167

Skeleton edges

8813 m

Skeleton length

80.34

Plan03 UDS

69.1%

Permeable

25.2%

Green pen.